Duke University **September 8, 2020**
Math 105L: Laboratory Calculus and Functions I

Big Assignment 1

- This assignment covers work from Day 1-2 to Day 4-2.
- You have ten days to complete the assignment. It is due Sept 18 at 8am.
- You may use the worksheets, labs, and your notes, as well as any other resources.
- However, you may not work with anyone else on this assignment – it must be entirely your own work.
- Throughout the assignment, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers. For example, copying graphs from graphing software is insufficient 'work'.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- You must hand in a scanned version of this assignment on Gradescope and no other pages. If you need to do rough work, do it elsewhere before writing your answers.
- You must sign the statement at the bottom of this page.

Honor Code:

I have adhered to the Duke Community Standard and all the rules above in completing this assignment.

Signature: _____

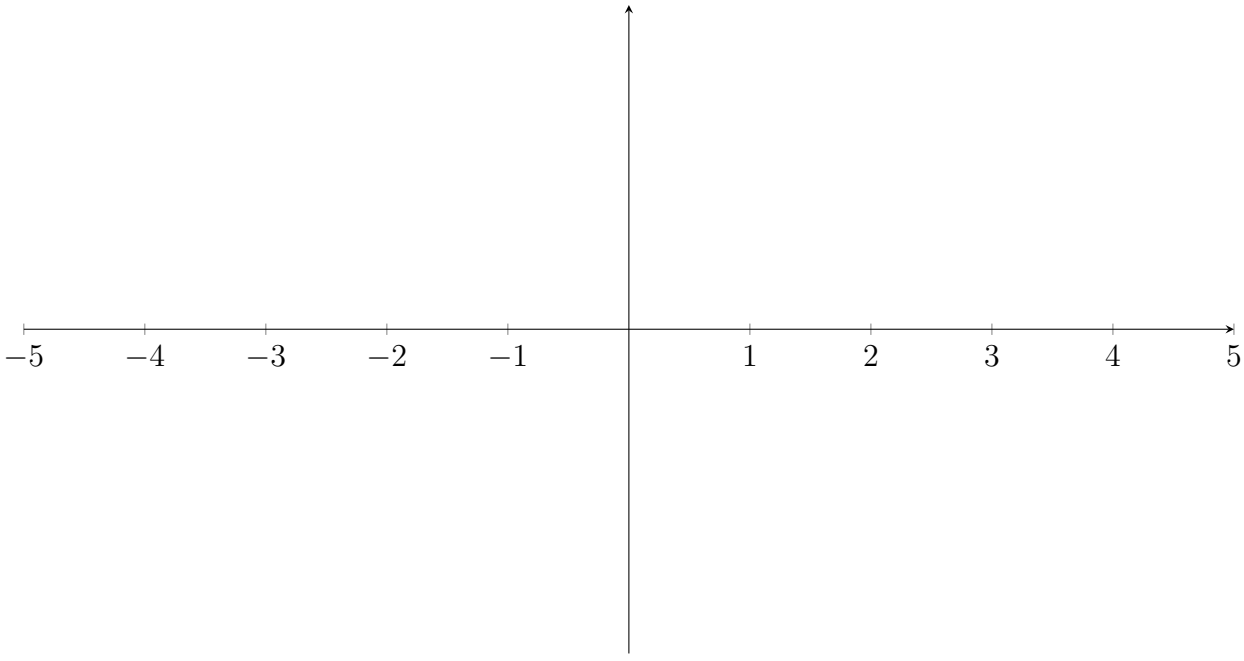
1. (10 points) Suppose that cats are afraid of water to varying degrees, depending on the depth of the water. The number of cats in the bathtub is given by $C = 10 - 2D$, where D is the depth of water in centimeters.

   (a) Write a short sentence describing the meaning of the number 10 in the equation.

   (b) Write a short sentence describing the meaning of the number -2 in the equation.

   (c) Suppose that the bathtub is in the shape of a box, with the bottom being 200cm by 60cm. Suppose also that the tap can fill the tub at the rate of $10,000$ cm^3 per minute. Lastly, suppose that you want to reduce the number of cats in the bathtub by 3 cats. How long will you have to run the tap for?

2. (12 points) Draw the graph of the function $f(x) = \frac{(x-1)(x^2+2x+1)}{2x^3+3x^2-2x}$ on the axes below, taking care to label important points and features. You must show all your work! (Hint: one factor of the denominator is $(x+2)$.)



3. (9 points)

(a) Find the equation of the linear function $f(x)$ that passes through $(1, 2)$ and $(6, -3)$.

(b) Let $g(x) = (x + 1)(x - 2)(x + 3)$. Find the x and y intercepts of $g(x)$, then draw the function.

(c) If the points $(1, 2)$, $(3, 4)$, and $(4, 3)$ are on the graph of a quadratic function $h(x)$, is the coefficient of x^2 positive or negative? Justify your answer briefly. (Hint: plot the points!)

4. (12 points) Let $f(x) = \dfrac{2x^2+3x+5}{3x^2+2x+1}$ and $g(x) = \dfrac{2x^2+3x-2}{3x^2+8x+4}$.

(a) Compute the following limits:

i. $\lim_{x\to 2} f(x)$

ii. $\lim_{x\to -2} g(x)$

(b) One of the two functions above is continuous on the real numbers. The other is not.

i. Which of the two is continuous?

ii. Find a function $h(x)$ that takes the same values as the discontinuous function above whereever both are defined, but is itself continuous. That is, find a function that 'fills in the hole'.

5. (15 points) Each of the following statements is false. For each one, explain why. You may draw graphs, write words, or provide any other valid explanation or counterexample.

   (a) Given the graph of $f(x)$, we can draw the graph of $f(2x+1)$ by first compressing the graph of $f(x)$ horizontally by a factor of 2, then shifting it left by 1.

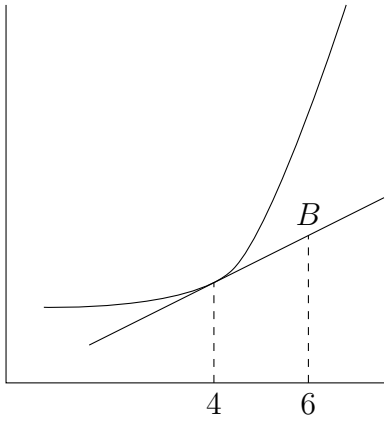
   (b) If the velocity of an object is negative, and its acceleration is positive, then the object is moving faster and faster as time goes by.

   (c) In an experiment, the height of an object at time $t = 0.03$ seconds is measured to be the same as its height at time 0 seconds. We can conclude that the object was dropped, rather than thrown upward or downward.

   (d) The expression $8|p| - 7g^2 = 9$ defines p as a function of g.

   (e) If $\lim_{x \to 7^+} f(x) = \lim_{x \to 7^-} f(x)$, then $f(x)$ is continuous at $x = 7$.

6. (13 points) Consider the following picture of a curve $f(x)$, a tangent line to the curve at the point $x = 4$, and a point B on the tangent line.



The equation of the tangent line (NOT THE CURVE) is given by

$$y = 6x - 3.$$

(a) Find the y-coordinate of the point B.

(b) Find the equation of the line that is perpendicular to the tangent line and that passes through B.

(c) Compute the following quantities and explain your answers:

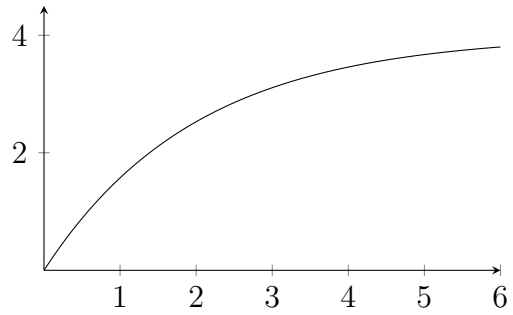
i. $f(4)$

ii. $f'(4)$

(d) Fill in the blank in the following expression with a greater than sign ($>$) or less than sign ($<$):

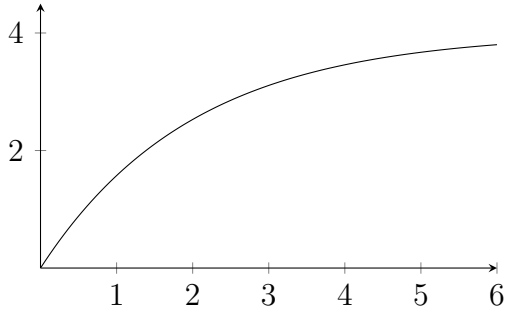
$$f'(6) \quad ___ \quad f'(4)$$

7. (10 points) The graph of a function $f(x)$ is shown below. Use it to answer the questions under the graph.



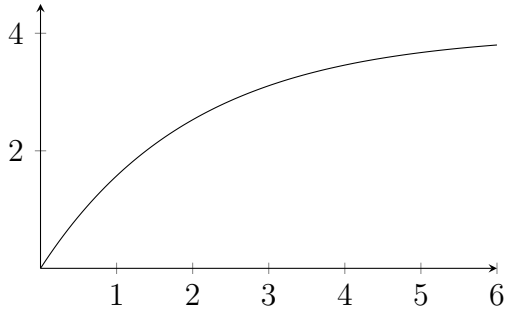
(a) Decide which of the following pairs is larger and fill in the blank with an inequality sign ($<$ or $>$). Use the copies of the graph to illustrate your answers. Carefully label your illustrations.

i. $A = f(2)$ and $B = f(4)$



A_____B

ii. A: The average rate of change of the function between $x = 2$ and $x = 4$, and B: the instantaneous rate of change of the function at $x = 4$.



A_____B

(b) What properties of the functions would you have to change in order to reverse each of the inequalities? Fill in the blanks below:

i. Right now, the function is _____. In order to reverse the first inequality, it would instead have to be _____.

ii. Right now, the function is _____ _____. In order to reverse the second inequality, it would instead have to be _____ _____.

8. (19 points)

(a) Complete the following definition of the derivative of a function at a point $x = a$ in its domain:

$$f'(a) = \lim_{_\to_} \text{_____}$$

(b) Consider the function $f(x) = 2x^2 + 3x$.

i. Using a table method, estimate the value of $f'(2)$.

ii. Using the limit definition of the derivative, show that $f'(2) = 11$. You may not use any derivative rules that we have not proved in class.

iii. Find the equation of the tangent line to $f(x)$ at $x = 2$.

iv. Draw a graph of $f(x)$ and its tangent line at $x = 2$. Label important points.

Bonus Question (2 points) Name all candidates for President of the United States on the NC ballot for 2020 and their parties.

Bonus Question (1 point) Tell me something interesting about yourself, or draw something awesome below.